$f: V \longrightarrow W$ $\quad$ 1) $f(u+v) = f(u) + f(v)$

$\quad\ ?\qquad\ ?$ $\quad$ 2) $f(\alpha v) = \alpha f(v) \quad \alpha \in \mathbb{R}$.

$f: \mathbb{R}^3 \to \mathbb{R}^2 \to f(x_1, x_2, x_3) = (x_1 - x_2, 2x_1 + x_3)$

$u = (u_1, u_2, u_3)$
$v = (v_1, v_2, v_3)$ $\Big\}$ $f(u+v) = f(u_1+v_1;\ u_2+v_2;\ u_3+v_3)$

$f(u+v) = (u_1+v_1 - (u_2+v_2),\ 2(u_1+v_1) + u_3+v_3)$

$f(u+v) = \underbrace{(u_1 - u_2;\ 2u_1+u_3)}_{f(u)} + \underbrace{(v_1 - v_2;\ 2v_1+v_3)}_{f(v)}$

$\alpha \in \mathbb{R}$

$v = (v_1, v_2, v_3) \Rightarrow f(\alpha v) = f(\alpha v_1, \alpha v_2, \alpha v_3)$

$f(\alpha v) = (\alpha v_1 - \alpha v_2;\ 2\alpha v_1 + \alpha v_3)$

$f(\alpha v) = \alpha\,(v_1 - v,\ 2v_1 + v_3)$

$f(\alpha v) = \alpha\, f(v)$

$f(-v) = -f(v)$

$f(\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3) = \alpha_1 f(v_1) + \alpha_2 f(v_2) + \alpha_3 f(v_3)$

$f: \mathbb{R}^2 \longrightarrow \mathbb{R}^3$

$\begin{cases} f(1,-1) = (1,0,-5) \\ f(0,2) = (-2,4,6) \end{cases}$ $\qquad B\{(1,-1)(0,2)\}$

$(x,y) = \alpha(1,-1) + \beta(0,2) = (\underset{x}{\alpha};\ \underset{y}{-\alpha+2\beta})$

$\alpha = x$

$y = -\alpha + 2\beta \longrightarrow \dfrac{y+x}{2} = \beta$

$f(x,y) = f\left[x(1,-1) + \dfrac{y+x}{2}(0,2)\right]$

$f(x,y) = x\, f(1,-1) + \dfrac{y+x}{2} f(0,2)$

$= x(1,0-5) + \dfrac{y+x}{2}(-2,4,6)$

$= (x - y - x,\ 2y + 2x,\ -5x + 3y + 3x)$

$= (-y,\ 2y + 2x,\ 3y - 2x)$

$s \in S \quad w = f(s) \to$ la imagen de S

$f^{-1}(w) \quad f(v) = w \longrightarrow$ preimagen de w.

t.L $f: \mathbb{R}^3 \to \mathbb{R}^3$ $\quad f(x,y,z) = (x+y-z;\ y+2z;\ x+2y+z)$

$S = \langle (1,2,0) \rangle \quad \bar{w} = (0,2,2) \quad W = \{ x \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0 \}$

Hallar $f(S)$; $f^{-1}(\bar{w})$; $f^{-1}(W)$

$S = \langle (1,2,0) \rangle \quad f(S) = \langle (3,2,5) \rangle$

$f^{-1}(\bar{w}) \to (x+y-z,\ y+2z,\ x+2y+z) = (0,2,2)$

$$\left(\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 2 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 2 & 2 \\ \cancel{0} & \cancel{-1} & \cancel{-2} & \cancel{-2} \end{array}\right) \to \begin{array}{l} y + 2z = 2 \to y = 2 - 2z \\ x + y - 2 = 0. \\ x = 2z - 2 - z \\ x = z - 2. \end{array}$$

$(x,y,z) = (z-2,\ 2-2z,\ z)$

$f^{-1}(\bar{w}) = z(1,-2,1) + (-2,2,0) \quad z \in \mathbb{R}$

$f^{-1}(W) \Rightarrow \bar{v} \in f^{-1}(W)$

$$\bar{v} = (x_1, x_2, x_3) \xrightarrow{tL} (\underbrace{x_1 + x_2 - x_3}_{x_1},\ \underbrace{x_2 + 2x_3}_{x_2},\ \underbrace{x_1 + 2x_2 + x_3}_{x_3}) = f(\bar{v})$$

$B \{ x_1 + x_2 + x_3 = 0 \} \Rightarrow x_1 + x_2 - x_3 + x_2 + 2x_3 + x_1 + 2x_2 + x_3 = 0.$

$2x_1 + 4x_2 + 2x_3 = 0.$

$f^{-}(W) = \{ \bar{x} \in \mathbb{R}^3 \mid 2x_1 + 4x_2 + 2x_3 = 0 \}.$

$Nu(f) = f^{-1}(\bar{0})$

$Im(f) = f(V) \xrightarrow[\bar{v} \in V]{} f(\bar{v}) = \bar{w}$

$f(x,y,z) = (x+2y-z;\ y+z;\ -x-y+2z)$

$f: \mathbb{R}^3 \to \mathbb{R}^3$

1) Hallar núcleo. $\begin{cases} x + 2y - z = 0 \\ y + z = 0 \\ -x - y + 2z = 0. \end{cases}$ $\ldots$ $\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0. \end{array}\right)$ $\quad \begin{array}{l} y + z = 0 \\ x + 2y - z = 0 \\ x = z - 2y \\ x = z - 2z \\ x = -z \end{array}$

$(x,y,z) = (-z, z, z) = z(-1,-1,1).$

Base $Nu(f) = \{(-1,1,1)\}.$

$\dim Nu f = 1.$

Imagen $\rightarrow$ $B\{(0,0,1)\ (0,1,0)\ (1,0,0)\}$. $\quad f(x,y,z) = (x+2y-z,\ y+z,\ -x-y+2z)$

tl (

$$Im(f) = \langle f(001), f(010), f(100)\rangle = \langle (1\,0\,-1)\ (2\,1\,-1)\ (-1\,1\,2)\rangle.$$

$$\begin{matrix} 1 & 0 & -1 \\ 2 & 1 & -1 \\ -1 & 1 & 2 \end{matrix} \rightarrow \begin{matrix} 1 & 0 & -1 \\ 0 & -1 & -3 \\ 0 & 1 & 1 \end{matrix} \rightarrow \begin{matrix} 1 & 0 & -1 \\ 0 & -1 & -3 \\ 0 & 0 & -2 \end{matrix} \rightarrow \begin{matrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{matrix}.$$

$$\alpha(1\,0\,1) + \beta(2\,1\,-1) + \gamma(-1\,1\,2) = (0,0,0).$$

$$\alpha + 2\beta - \gamma = 0.$$
$$\beta + \gamma = 0.$$
$$\alpha - \beta + 2\gamma = 0.$$

$B_{Im(f)} = \{(1\,0\,-1)\ (2\,1\,-1)\}$

$Dim\ Im(f) = 2.$

$f: V \rightarrow W \qquad f$: BO

$$\bar{v} \in V \rightarrow \{v_1, v_2, \dots v_n\} \xrightarrow{LI} \{f(v_1), f(v_2) \dots\} \in W$$

1. $f: \mathbb{R}^3 \rightarrow \mathbb{R}^6$ monomorfismo $\rightarrow Nu = \{\bar{0}\} \rightarrow \dim Nu = 0$

   $\dim Im(f) = 3.$

2. $f: \mathbb{R}^5 \rightarrow \mathbb{R}^3$ epimorfismo. $\rightarrow Im(f) = W \leftarrow$ espacio llegada.

   $\dim(Im(f)) = 3 \rightarrow \dim Nu(f) = 2.$

3. $f: V \rightarrow V$

   $f(v_1) = v_1 + v_2$
   $f(v_2) = -v_2 + v_3$
   $f(v_3) = v_1 + 2v_3.$

   $\{v_1, v_2, v_3\}$ Base de V

   $\dim V = 3.$

$$Im\ f = \langle f(v_1), f(v_2), f(v_3)\rangle = \langle v_1+v_2, -v_2+v_3, v_1+2v_3\rangle$$

$(v_1+v_2)_B = (1,1,0) \qquad (-v_2+v_3)_B = (0,-1,1) \qquad (v_1+2v_3)_B = (1\,0\,2)$

$Im(f) = \{(1\,1\,0)\ (0\,-1\,1)\ (1\,0\,2)\}$ LI $\rightarrow$ Base

$\dim Im(f) = 3 \rightarrow \dim(Nu) = 0.$

composición $f \circ g : V \to W$ $\quad (g \circ f)\bar{v} = g(f(\bar{v}))$

$g(f(\bar{v}+\bar{w})) = g(f(\bar{v}) + f(\bar{w})) = g(f(\bar{v})) + g(f(\bar{w}))$

$g(\alpha f(\bar{v})) = \alpha\, g(f(\bar{v}))$

$\| \; Nu(f) \subseteq Nu(g \circ f) \; \|$

$\| \; Im(g \circ f) \subseteq Im(g) \; \|$

Ej: $f(1-11) = (000)$

$f(-110) = (211)$

$f(100) = (11,-2)$

Base $\{(1-11)\ (-110)\ (100)\}$.

$f: \mathbb{R}^3 \to \mathbb{R}^3$.

$g(x,y,z) = (x-y+2,\ -x+y-2)$

$g: \mathbb{R}^3 \to \mathbb{R}^2$.

Hallar $Im(g \circ f)$ $\quad Nu(g \circ f)$

$Im(g)$ $\quad Nu(f)$

$g \circ f = g(f(\bar{x})) \to$ $g(f(1-11)) = g(000) = (0,0)$

$g(f(-110)) = g(211) = (2,-2)$

$g(f(100)) = g(11-2) = (-2,2)$

$Im(g \circ f) = \langle (0,0)\ (2-2)\ (-2,2) \rangle$

$B\ Im(g \circ f) = \{(-22)\}$. $\quad \dim Im(g \circ f) = 1$.

$\dim Nu(g \circ f) = 2$.

$g(f(-110) + f(100)) = (2-2) + (-22)$

$g(f(010)) = (0,0)$

$B\ Nu(g \circ f) = \{(1-11)\ (010)\}$.

$Im\, g(\bar{x}) = \langle g(100),\ g(010),\ g(001) \rangle = \langle (11,-1) \cdot (-11)(1-1) \rangle$

$B\ Im(g(\bar{x})) = \{(1,-1)\}$ $\dim Img = 1$.

$Nu\, f \Rightarrow$ $Im\, f = \langle f(1-11),\ f(-110),\ f(100) \rangle = \langle (000), (211)\ (11-2) \rangle$

$\dim Im(f) = 2$.

$\dim Nu(f) = 1$

$B\ Nu\, f = \{(1-11)\}$

Inversa TL.

$f: V \to W$ isomorfismo.

$f^{-1}: W \to V \to f \circ f^{-1} = id_W. \quad f^{-1} \circ f = id_V.$

$f(f^{-1})$

$f(x,y,z) = (x-y+z, y-z, 3x+y+z)$

$f = \mathbb{R}^3 \to \mathbb{R}^3$

$B\{(100)\ (010)\ (001)\} \xrightarrow{tl} f(100) = (103) \quad f(010) = (-111) \quad f(001) = (1-11)$

$B\{(103)\ (-111)\ (1-11)\}$.

$f^{-1} \circ f = id_{\mathbb{R}^3}$

$f^{-1}(f(100)) = f^{-1}(103) = (100)$

$f^{-1}(f(010)) = f^{-1}(-111) = (010)$

$f^{-1}(f(001)) = f^{-1}(1-11) = (001)$

Proyector

$f: V \to V \quad p \circ p = p.$

$p(x,y) = \left(\frac{x+y}{2}, \frac{x+y}{2}\right) \to p \circ p = p(p(x,y))$

$p(p(x,y)) = p\left(\frac{x+y}{2}, \frac{x+y}{2}\right) = \left(\frac{\frac{x+y}{2} + \frac{x+y}{2}}{2}, \frac{\frac{x+y}{2} + \frac{x+y}{2}}{2}\right) = \left(\frac{x+y}{2} + \frac{x+y}{2}\right)$

$\bar{v} \in Im\ p \Rightarrow p(\bar{v}) = \bar{v}$

$V = Nu(p) \oplus Im(p)$

$f: \mathbb{R}^3 \to \mathbb{R}^3.$

$Nu\langle p \rangle = \langle (110) \rangle \qquad Im(p) = \{x \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 0\}.$

$p(110) = (000) \qquad Im(p) = \langle (-110)\ (-101) \rangle$

$B = \{(-110)\ (-101)\ (110)\}$

$p(-110) = (-110)$

$p(-101) = (-101)$

$p(110) = (000)$.

$p \circ p = p \to p(p(-110)) = p(-110)$

$p(p(-101)) = p(-101)$

$p(p(110)) = p(000) = (0,0,0) = p(110)$